

Customer Lifetime Value Prediction Using Machine Learning

1. Introduction

Customer Lifetime Value (CLV) is a metric that estimates the total revenue a customer is expected to generate throughout their relationship with a business. Predicting CLV helps organizations identify valuable customers, improve retention strategies, and optimize marketing expenditure.

The objective of this project is to develop a machine learning model that predicts Customer Lifetime Value using customer purchasing behavior from an online retail dataset.

2. Objectives

- To analyze customer transaction data.
- To calculate customer-level purchasing metrics.
- To build a machine learning model for CLV prediction.
- To evaluate the performance of the model.
- To provide business insights for customer management.

3. Dataset Description

Dataset Used: Online Retail Dataset

Attributes:

- Invoice Number
- Stock Code
- Product Description
- Quantity
- Invoice Date
- Unit Price
- Customer ID
- Country

The dataset contains transaction records of customers purchasing products from an online retail store.

4. Data Preprocessing

The following preprocessing steps were performed:

1. Removed missing Customer IDs.
2. Removed cancelled transactions.
3. Removed records with negative quantities.
4. Removed records with negative prices.
5. Created Revenue = Quantity $\times$ Unit Price.

5. Feature Engineering

Customer-level features were created:

- Frequency: Number of unique purchases.
- TotalQuantity: Total products purchased.
- AvgOrderValue: Average value per order.
- CLV: Total revenue generated by the customer.

6. Machine Learning Model

Algorithm Used:

Random Forest Regressor

Reason for Selection:

- Handles nonlinear relationships.
- Provides high predictive accuracy.
- Reduces overfitting through ensemble learning.

7. Model Evaluation

Evaluation Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Results:

- MAE = 454.31
- RMSE = 5091.47
- $R^2 = 0.747$

The R^2 score indicates that the model explains approximately 75% of the variation in Customer Lifetime Value.

8. Visualizations

- CLV Distribution
- Feature Importance Plot
- Actual vs Predicted Plot
- Residual Distribution Plot

9. Business Insights

- Customers with higher purchase frequency tend to generate higher CLV.
- Average order value significantly influences customer value.
- High-value customers can be targeted with loyalty programs.
- CLV prediction supports revenue forecasting and marketing optimization.

10. Conclusion

The project successfully developed a machine learning model for Customer Lifetime Value prediction using Random Forest Regression. The model demonstrated good predictive performance and can assist businesses in identifying valuable customers and making data-driven decisions.

